

Appendix: AI Protocols, Diagrammatic Reasoning, and Notation Conventions

H. AI and Communication Protocols

Term	Symbol	Definition	First Use
A2A Protocol	—	Google's Agent-to-Agent protocol for cross-framework agent communication via HTTP/JSON-RPC	§9
ACP	—	Agent Communication Protocol; standardizes messaging formats	§9

I. Diagrammatic Reasoning

Term	Symbol	Definition	First Use
Cognitively privileged representation	—	Representation format that leverages perceptual and spatial cognition for inference	§1
Diagram depth	—	Length of the longest input-to-output path through boxes; measures derivational complexity	§4
Existential	—	Peirce's	§7

J. Notation Conventions

Convention	Meaning
$\mathcal{C}, \mathcal{D}$	Categories
$\mathcal{V}$	Enrichment base (monoidal category, typically $([0, 1], \cdot, 1)$)
$\mathcal{U}$	Universal (maximal) case category
$\mathcal{L}$	Language-specific case category
$\mathcal{E}_{\mathbb{T}}$	Classifying topos of theory $\mathbb{T}$
f, g, h	Morphisms
F, G	Functors
α, β	Natural transformations
n, s	Basic pregroup types: noun, sentence
n^l, n^r	Left and right adjoints of type n
$n_{\text{NOM}}, n_{\text{ACC}}$	Case-subscripted noun types (e.g., nominative noun, accusative noun)
N, S	Noun space and sentence space